Chapter 8. Causal Data Science

Transportability, selection bias, data fusion, missing data

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Motivation. How to answer a causal query about a target population from data collected under different conditions, populations, sampling, missingness or measurements.

8.1 Motivation and big picture

Motivation. Real data rarely match the inferential target; the mismatches fall into a few dimensions.

Examples. *Genetics:* carcinogenic in rats, hence in humans? *Advertisement:* what transfers from product A's campaign to product B? *Robotics:* how can a rover trained in the California desert use Earth experience on Mars?

"Not all data are created equal." Data are collected under conditions differing from the target:

(1) different experimental conditions, (2) different populations, (3) non-random sampling, (4) non-random treatment assignment, (5) unmeasured relevant variables. These dimensions have been formalized, with conditions and algorithms.

Heterogeneous data sets. For $Q = P(y \mid \text{do}(x))$, data sets may differ along four dimensions:

		Dataset 1	Dataset 2	Dataset n
d_1	population	Los Angeles	New York	Seoul
d_2	obs./exp.	experimental (randomized Z_1)	observational	experimental (randomized Z_2)
d_3	sampling	selection on age	selection on SES	–
d_4	measured	$\{X_1, Z_1, W, M, Y_1\}$	$\{X_1, X_2, Z_1, N, Y_2\}$	$\{X_2, Z_1, W, L, M, Y_1\}$

Goal. Make the most of multiple heterogeneous data sets with formal machinery (language, conditions, algorithms) for causal inference and *fusion* across studies.

Classic identifiability takes a query $P_x(y) = P(y \mid \text{do}(x))$, a diagram and $P(\mathbf{V})$, and returns "no" or $P_x(y) = f_{\mathcal{G}}(P(\mathbf{V}))$. In data fusion the target may be the US with sources NY, LA, Boston, ..., each with its own diagram $(\mathcal{G}_{US}, \mathcal{G}_{NYC}, \dots)$ and distribution $(P_{NYC} + P_{LA} + \cdots)$:

(a) US: census data available (the target population);

(b) NY: survey data resembling the target;

(c) LA: survey data, younger subjects;

(d) Boston: age not recorded, mostly successful lawyers;

(e) SF: high post-treatment blood pressure;

(f) Texas: mostly Spanish subjects, high attrition;

(g) Arkansas: randomized trial on college students;

(h) Utah: RCT with paid volunteers, mainly unemployed;

(i) Wyoming: natural experiment on young athletes.

Tasks. With data collections as tuples (d_1, d_2, d_3, d_4):

1. Descriptive statistics: $(d_1, \text{Samples(Obs)}, d_3, d_4) \to (d_1, \text{Distribution(Obs)}, d_3, d_4)$.
2. Experimental statistics: $(d_1, \text{Samples}(\text{do}(x)), d_3, d_4) \to (d_1, \text{Distribution}(\text{do}(x)), d_3, d_4)$.
3. Causal inference from observational studies: $(d_1, \text{Obs}, d_3, d_4) \to (d_1, \text{do}(x), d_3, d_4)$ (mainly confounding bias, 1974–2011: Rubin, Robins, Dawid, Pearl, ...).
4. Generalized identification (experiments on other variables): $(d_1, \text{do}(Z), d_3, d_4) \to (d_1, \text{do}(x), d_3, d_4)$.

5. Sampling selection bias: $(d_1, d_2, \text{Select}(\text{Age}), d_4) \to (d_1, d_2, \{\}, d_4)$.
6. Transportability (external validity): $(\text{Bonobos}, d_2, d_3, d_4) \to (\text{Humans}, d_2, d_3, d_4)$.

Usually treated in isolation under parametric assumptions; together they form "causal data science".

Four dimensions. (1) Experimental conditions: generalized identification.

(2) Environmental conditions: transportability.

(3) Sampling conditions: recovering from selection bias.

(4) Responding conditions: recovering from missingness. Plus measurement (partial observation).

8.2 Experimental conditions: general identifiability

Motivation. Identify an effect from observational data plus experiments on other variables.

8.2.1 The challenge

Diet Z, cholesterol X, heart attack Y: $Z \to X \to Y$, $Z \leftrightarrow X$, $Z \leftrightarrow Y$. Given $P(x, y, z)$ and $P(x, y \mid \text{do}(z))$:

$$Q = P(y \mid \text{do}(x)) = \frac{P(x, y \mid \text{do}(z))}{P(x \mid \text{do}(z))} = P(y \mid \text{do}(z), x)$$

(Rule 3 adds $\text{do}(z)$; Rule 2 exchanges $\text{do}(x)$ for x, as $\mathcal{G}_{\overline{Z}\,\underline{X}}$ has no back-door path). From $P(x, y, z)$ alone Q is not identifiable ($X \leftrightarrow Z \leftrightarrow Y$), so the experiment is essential.

8.2.2 Subtleties of z-ID

Same query, three graphs ($\mathcal{G}$ with $P(x, y, z)$; $\mathcal{G}_{\overline{Z}}$ with $P(x, y \mid \text{do}(z))$; $\mathcal{G}_{\overline{X}}$ for the query):

- $Z \to X \to Y$, $Z \leftrightarrow X$, $Z \leftrightarrow Y$: **z-ID**.
- $Z \to X \to Y$, $X \leftrightarrow Y$, $Z \leftrightarrow Y$: **not z-ID**; $X \leftrightarrow Y$ persists in $\mathcal{G}_{\overline{Z}}$ (linear IV would help, nonparametric IV does not).
- $Z \leftrightarrow X$, $Z \to Y$, $Z \leftrightarrow Y$, $X \to Y$: **not z-ID**; Z does not cause X.

8.2.3 z-ID reduced to calculus

$Z \to X \to W \to Y$, $Z \leftrightarrow X$, $Z \leftrightarrow W$, $Z \leftrightarrow Y$, $X \leftrightarrow Y$, with $P(\mathbf{v})$ and $P(\mathbf{v} \mid \text{do}(z))$:

$$\begin{aligned} P(y \mid \text{do}(x)) &= \sum_w P(y \mid \text{do}(x), w) P(w \mid \text{do}(x)) = \sum_w P(y \mid \text{do}(x, w)) P(w \mid \text{do}(x)) \\ &= \sum_w \underbrace{P(y \mid \text{do}(w))}_{Q[Y]} \underbrace{P(w \mid \text{do}(x))}_{Q[W]}. \end{aligned}$$

Neither factor is identifiable from $P(\mathbf{V})$; both are from $P(\mathbf{v} \mid \text{do}(z))$:

$$\begin{aligned} Q[Y] &= P(y \mid \text{do}(w)) = P(y \mid \text{do}(w, z)) = \sum_x P(y \mid \text{do}(w, z), x) P(x \mid \text{do}(w, z)) \\ &= \sum_x P(y \mid \text{do}(w, z), x) P(x \mid \text{do}(z)) = \sum_x P(y \mid \text{do}(z), w, x) P(x \mid \text{do}(z)), \\ Q[W] &= P(w \mid \text{do}(x)) = P(w \mid \text{do}(x, z)) = P(w \mid \text{do}(z), x). \end{aligned}$$

8.2.4 Example: drug–drug interactions

Y: cardiovascular disease, B: blood pressure, X_1: antihypertensive, X_2: anti-diabetic. Graph (a): $X_1 \to B \to Y$, $X_2 \to Y$, $X_1 \leftrightarrow B$, $X_1 \leftrightarrow X_2$, $X_2 \leftrightarrow Y$. Effect of *both* drugs from single-drug experiments $\mathbb{Z} = \{\{X_1\}, \{X_2\}\}$:

$$P_{x_1,x_2}(y) = \sum_b P_{x_1,x_2}(y,b) = \sum_b P_{x_1}(b) P_{x_2,b}(y) = \sum_b P_{x_1}(b) P_{x_2}(y \mid b),$$

$P_{x_1}(b)$ from the X_1 trial, $P_{x_2}(y \mid b)$ from the X_2 trial (B unconfounded with Y in $\mathcal{M}_{x_2}$). Adding $B \leftrightarrow X_2$ (b) keeps identifiability; $B \leftrightarrow Y$ (c) or $X_1 \leftrightarrow Y$ (d) destroys it.

8.2.5 Algorithm and summary

Two phases, using probability axioms and do-calculus:

(1) modify and decompose the query into factors;

(2) identify each factor from some $P_{\mathbf{Z}_j} \in \mathbb{P} = \{P_{\mathbf{Z}_1}, \ldots, P_{\mathbf{Z}_m}\}$; fail only if some factor is identifiable from none.

Identifiability of any $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{z})$ from observational plus experimental studies is decidable for any graph, estimands are found in polynomial time, and do-calculus is complete. References: Pearl (1993, 1994); Tian and Pearl (2002); Shpitser and Pearl, Huang and Valtorta (2006); Bareinboim and Pearl (2012) z-ID; Lee, Correa and Bareinboim (2019) g-ID; Correa and Bareinboim (2019, 2020) σ-calculus.

8.3 Environmental conditions: transportability

Motivation. Transfer causal knowledge from source populations to a different target.

8.3.1 The problem

Can the effect of $\mathbf{X}$ on $\mathbf{Y}$ in a target Π^* be computed from findings in other populations (e.g. US education policy applied to South Korea)? Elsewhere this is external validity (Shadish, Cook and Campbell 2002; Cox 1958 asks for "some understanding of the reasons for the differences"; Manski 2007). Internal validity concerns the study's own population.

From lab to real world. H_0: all mechanisms agree ($f_z = f_z^*, \ldots$), trivially transportable; H_a: all differ, nothing transports. Real problems lie between.

8.3.2 What can an experiment in LA tell us about NYC?

Z (age) $\to X$, $Z \to Y$, $X \to Y$; only the age distribution differs. LA: $P(x,y,z)$ and $P(y \mid \mathrm{do}(x), z)$; NYC: $P^*(x,y,z)$. Then

$$Q = P^*(y \mid \mathrm{do}(x)) = \sum_z P(y \mid \mathrm{do}(x), z)\, P^*(z)\,:$$

LA's z-specific effects reweighted by NYC's ages.

Transport formulas depend on the causal story. (a) Z = age: $P^*(y \mid \mathrm{do}(x)) = \sum_z P(y \mid \mathrm{do}(x), z) P^*(z)$.

(b) Z = language skill, a causally irrelevant proxy: $P^*(y \mid \mathrm{do}(x)) = P(y \mid \mathrm{do}(x))$.

(c) Z = bio-marker on $X \to Z \to Y$: $P^*(y \mid \mathrm{do}(x)) = \sum_z P(y \mid \mathrm{do}(x), z) P^*(z \mid x)$. ($P^*$: target, P: source.)

8.3.3 Selection diagrams

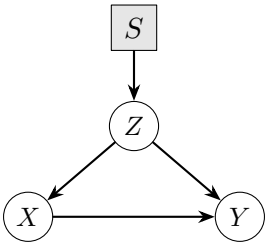


Figure. *Selection diagram for LA/NYC: $S \to Z$ marks the only differing mechanism, so* $P^*(y \mid \mathrm{do}(x)) = \sum_z P(y \mid \mathrm{do}(x), z)P^*(z)$.

Overlay source $\mathcal{G}$ and target $\mathcal{G}^*$ and add a *selection node* S (small square) into each variable whose mechanism or exogenous distribution may differ. E.g. $\mathcal{G}$: $Z \to X$, $Z \to Y$, $X \to W \to Y$, $\mathcal{G}^*$ without $Z \to X$; with $f_X \neq f_X^*$, $f_Y \neq f_Y^*$ and the rest equal, $S \to X$, $S \to Y$ (edges present in only one graph: use the union). Target queries: $P^*(\cdot) = P(\cdot \mid S = s^*)$.

Theorem 8.1 (Transportability; Pearl and Bareinboim 2011)

Q is transportable from Π to Π^* iff there is a do-calculus reduction of $Q(\Pi^*)$ to an estimand that is a function of the observed distributions.

Proof [Supplement]. *If:* the selection diagram is one causal model with nodes S, and $Q(\Pi^*) = P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), S = s^*)$. Each do-calculus step is valid there, so a reduction to observables (with S only in factors estimable in Π^*) expresses $Q(\Pi^*)$ through available data. *Only if:* Bareinboim and Pearl (2012) construct, for non-transportable queries, two pairs of domains agreeing on all available distributions but not on $Q(\Pi^*)$; not reproduced here. ■

Example 8.1

$S \to Z$, $Z \to X$, $Z \to W \to Y$, $X \to Y$, $Z \leftrightarrow X$, $Z \leftrightarrow W$, $X \leftrightarrow Y$:

$$\begin{aligned} Q = P^*(y \mid \mathrm{do}(x)) &= P(y \mid \mathrm{do}(x), S) = \sum_w P(y \mid \mathrm{do}(x), S, w)P(w \mid \mathrm{do}(x), S) \\ &= \sum_w P(y \mid \mathrm{do}(x), w)P(w \mid \mathrm{do}(x), S) = \sum_w P(y \mid \mathrm{do}(x), w)P(w \mid S) \\ &= \sum_w P(y \mid \mathrm{do}(x), w)P^*(w). \end{aligned}$$

Step 3: $(Y \perp_d S \mid X, W)_{\mathcal{G}_{\overline{X}}}$ (Rule 1); step 4: $X \notin An(W)$ (Rule 3). First factor from the source experiment, second from target observations.

Algorithm. Input: a selection diagram. Output: transportability and, if so, a formula saying what to measure where, e.g.

$$P^*(y \mid \mathrm{do}(x)) = \sum_z P(y \mid \mathrm{do}(x), z) \sum_w P^*(z \mid w) \sum_t P(w \mid \mathrm{do}(x), t)P^*(t).$$

A c-factor not identifiable from $P^*(\mathbf{V})$ must come from source experiments, possible only if no selection node points into it.

8.3.4 General transportability

Experiments over $\{\mathbf{Z}^1, \ldots, \mathbf{Z}^n, \mathbf{Z}^*\}$ in domains $\Pi = \{\Pi^1, \ldots, \Pi^n, \Pi^*\}$ sometimes suffice; the theory says when and how.

Example 8.2

$Q = P^*(y \mid \text{do}(x))$ from Π^a (LA; experiments on X) and Π^b (NYC; experiments on Z). Both: $X \to Z \to Y$, $X \leftrightarrow Z$, $X \leftrightarrow Y$; selection nodes into X, Y (Π^a) and X, Z (Π^b). Neither source alone suffices. Combined (S_{xzy}: selection nodes),

$$
\begin{aligned}
Q = P(y \mid \text{do}(x), S_{xzy}) &= \sum_z P(y \mid \text{do}(x), z, S_{xzy}) P(z \mid \text{do}(x), S_{xzy}) && \text{def., axioms} \\
&= \sum_z P(y \mid \text{do}(x), \text{do}(z), S_{xzy}) P(z \mid \text{do}(x), S_{xzy}) && \text{Rule 2} \\
&= \sum_z P(y \mid \text{do}(z), S_{xzy}) P(z \mid \text{do}(x), S_{xzy}) && \text{Rule 3} \\
&= \sum_z P(y \mid \text{do}(z), S_y) P(z \mid \text{do}(x), S_z) && \text{Rule 1} \\
&= \sum_z \underbrace{P^{(b)}(y \mid \text{do}(z))}_{\text{NYC}} \underbrace{P^{(a)}(z \mid \text{do}(x))}_{\text{LA}} && \text{def.}
\end{aligned}
$$

($P(y \mid \text{do}(z))$ from where S does not point to Y, $P(z \mid \text{do}(x))$ from where it does not point to Z; selection diagrams extend to multiple sources.)

8.3.5 Is the gold standard golden?

$Z \to X$, $Z \to Y$, $X \to Y$ with confounding and $S \to Z$. Randomization removes edges into X, but $S \to Z \to Y$ remains: even a perfect RCT needs a transportability argument.

Summary. Nonparametric transportability is decidable from selection diagrams, formulas are derived in polynomial time, and calculus and algorithm are complete, addressing external validity, meta-analysis and heterogeneity. References: Pearl and Bareinboim (2011); Bareinboim and Pearl (2012, 2013a,b, 2014); Lee, Correa and Bareinboim (2020).

8.4 Sampling conditions: recovering from selection bias

Motivation. Recover population quantities from samples whose inclusion depends on the variables.

8.4.1 The problem

Preferential inclusion of samples biases inference. Add a selection indicator S ($S = 1$: sampled). In Pearl's seven-variable model ($Z_1 \to W_1 \to X$, $Z_1 \to Z_3$, $Z_2 \to Z_3$, $Z_2 \to W_2 \to Y$, $Z_3 \to X$, $Z_3 \to Y$, $X \to W_3 \to Y$), isolated S is random sampling; $W_1 \to S$, $W_2 \to S$ make it covariate-dependent. When is, e.g., $P(y \mid x)$ estimable from $P(\mathbf{v} \mid S = 1)$?

8.4.2 The origin of selection bias

$X = U_x$, $Y = U_y$, $S = \neg(X \oplus Y)$, $P(u_x, u_y) = 1/4$. In the sample $P(X = 1, Y = 1 \mid S = 1) = P(X = 0, Y = 0 \mid S = 1) = 1/2$:

$$P(Y = 1 \mid X = 1, S = 1) - P(Y = 1 \mid X = 0, S = 1) = 1,$$

while in the population the difference is 0. Conditioning on a common effect ($X \to S \leftarrow Y$) creates dependence.

Confounding versus selection bias. Both are unblockable spurious flows: through a common cause $X \leftarrow Z \rightarrow Y$, or through a conditioned common effect $X \rightarrow S \leftarrow Y$.

Transportability versus selection. Nature-made versus man-made: in transportability S marks difference-producing factors; in selection its parents are sample-producing factors.

8.4.3 Variants

In statistics:

(i) without external data, is the odds ratio $OR(y,x) = \frac{P(y|x)P(y'|x')}{P(y'|x)P(y|x')}$ recoverable from $P(\mathbf{v} \mid S=1)$? (ii) without external data, is $P(y \mid x)$ recoverable? (iii) with external $P(\mathbf{t})$, $\mathbf{T} \subseteq \mathbf{V}$, is $P(y \mid x)$ recoverable? In causal inference: is $P(y \mid \text{do}(x))$ recoverable from $P(\mathbf{v} \mid S=1)$ and $P(\mathbf{t})$, or, with instruments, bounds on it?

8.4.4 Recoverability without external data

Theorem 8.2 (Bareinboim, Tian and Pearl 2014)

$Q = P(y \mid x)$ is recoverable from selection-biased data iff $(S \perp\!\!\!\perp Y \mid X)$.

Proof [Supplement]. *If:* $P(y \mid x) = P(y \mid x, S=1)$, estimable from the sample. *Only if:* if $S \not\perp\!\!\!\perp Y \mid X$, Bareinboim, Tian and Pearl (2014) construct two models compatible with $\mathcal{G}$ agreeing on $P(x,y \mid S=1)$ but not on $P(y \mid x)$; not reproduced here. ■

$Y \rightarrow S$: not recoverable; only $X \rightarrow S$: $P(y \mid x) = P(y \mid x, S=1)$.

8.4.5 Recoverability with external data

With $W_1 \rightarrow S \leftarrow W_2$, $P(y \mid x)$ is not recoverable from the sample alone; what if unbiased $P(W_1, W_2)$ is available?

Theorem 8.3 (Sufficiency)

$P(y \mid x)$ is recoverable if there is a set C with $(Y \perp\!\!\!\perp S \mid C, X)$ in $\mathcal{G}$ and $P(C, X)$ estimable; then

$$P(y \mid x) = \sum_c P(y \mid x, c, S=1) P(c \mid x).$$

Proof [Supplement]. $P(y \mid x) = \sum_c P(y \mid x, c) P(c \mid x)$, and $(Y \perp\!\!\!\perp S \mid C, X)$ gives $P(y \mid x, c) = P(y \mid x, c, S=1)$, estimable; $P(c \mid x)$ is available by assumption. ■

$C = \{W_1, W_2\}$: yes (parents of S). $C = \{W_1, Z_1, Z_2\}$: no ($S \leftarrow W_2 \rightarrow Y$ open). $C = \{W_2, Z_3\}$: yes.

Confounding and selection together (C given externally):

(1) $W_1 \rightarrow X$, $W_1 \rightarrow W_2 \rightarrow Y$, $X \rightarrow Y$, $W_2 \rightarrow S$: $C = \{W_2\}$ recovers $P(y \mid \text{do}(x))$; $C = \{W_1\}$ does not.

(2) $X \rightarrow Y$, $X \rightarrow W \leftarrow Y$, $W \rightarrow S$: $C = \{\}$ and $C = \{W\}$ fail, $C = \{W, X\}$ works.

Theorem 8.4 (Generalized adjustment; Correa, Tian and Bareinboim 2018)

$(\mathbf{Z}, \mathbf{Z}^T)$ with $\mathbf{Z}^T \subseteq \mathbf{Z}$ (the part measured without bias) is admissible for generalized adjustment relative to $(\mathbf{X}, \mathbf{Y})$ iff

1. $\mathbf{Z}$ does not disturb any proper causal path;
2. $\mathbf{Z} \cup \{S\}$ blocks all non-causal paths (cut the first edge of every proper causal path; then $\mathbf{Z} \cup \{S\}$ must separate $\mathbf{X}$ and $\mathbf{Y}$);
3. $\mathbf{Z}^T$ d-separates $\mathbf{Y}$ from S in the graph where proper causal paths have been severed.

Then

$$P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{z}} \underbrace{P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}, S=1) P(\mathbf{z} \setminus \mathbf{z}^T \mid \mathbf{z}^T, S=1)}_{\text{biased data}} \underbrace{P(\mathbf{z}^T)}_{\text{unbiased data}} .$$

Proof [Supplement]. *Sufficiency.* Condition 3 gives $\mathbf{Y} \perp\!\!\!\perp S \mid \mathbf{X}, \mathbf{Z}$ in the relevant graph, so $P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}) = P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}, S=1)$. Conditions 1–2 make $\mathbf{Z}$ an adjustment set: $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{z}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}) P(\mathbf{z})$. Finally $P(\mathbf{z}) = P(\mathbf{z} \setminus \mathbf{z}^T \mid \mathbf{z}^T) P(\mathbf{z}^T)$ with $P(\mathbf{z}^T)$ unbiased and $P(\mathbf{z} \setminus \mathbf{z}^T \mid \mathbf{z}^T) = P(\mathbf{z} \setminus \mathbf{z}^T \mid \mathbf{z}^T, S=1)$ by the remaining condition. *Necessity*: Correa, Tian and Bareinboim (2018), not reproduced here. ■

Example 8.3

With $\mathbf{Z} = \{Z_1, Z_2, Z_3\}$ satisfying the conditions and $\mathbf{Z}^T = \{Z_2, Z_3\}$: $P(y \mid \mathrm{do}(\mathbf{x})) = \sum_{z_1, z_2, z_3} P(y \mid \mathbf{x}, z_1, z_2, z_3, S=1) P(z_1 \mid z_2, z_3, S=1) P(z_2, z_3)$.

Summary. Recoverability is decidable from a diagram augmented with the selection mechanism, estimands are polynomial-time, results are complete for pure recoverability and sufficient with external data (Correa, Tian and Bareinboim 2019), and the back-door criterion generalizes to confounding plus selection. References: Bareinboim, Tian and Pearl (2014); Bareinboim and Tian (2015); Correa, Tian and Bareinboim (2018, 2019a,b).

8.5 Responding conditions: missing data

Motivation. Missingness graphs decide when quantities are recoverable from incomplete data.

(Partly following Mohan and Pearl 2021, "Graphical models for processing missing data", JASA.)

8.5.1 Motivation

Sensors fail, respondents skip questions, patients forget. The Rubin-rooted literature mostly assumes MAR, needed for EM (Dempster et al. 1977) and multiple imputation (Rubin 1978) to be consistent; yet MAR is almost impossible to verify, and beyond-MAR results are scarce. Graphs encode the *reasons* for missingness.

Why graphs? *Transparency:* MCAR/MAR independences are invisible in data but visible in the graph. *Recoverability:* decide whether and how any method can consistently estimate a parameter. *Testability:* check the model against the (corrupted) data via implied independences.

8.5.2 A motivating example

Age A, Gender G, Obesity O ($G \to O \leftarrow A$); A, G from school records, O missing (m) for students hiding their weight. The proxy is

$$O^* = f(R_O, O) = \begin{cases} O & R_O = 0, \\ m & R_O = 1. \end{cases}$$

Missingness may be random (lost questionnaires), caused by a fully observed variable (rebellious teenagers, $A \to R_O$), or by a partially observed one (embarrassed obese students, $O \to R_O$).

8.5.3 m-graphs

Nodes: $\mathbf{U} \cup \mathbf{V}_o \cup \mathbf{V}_m \cup \mathbf{V}^* \cup \mathbf{R}$, i.e. latents, always-observed variables, sometimes-missing variables, proxies (V_i^* observed for $V_i \in \mathbf{V}_m$), and missingness mechanisms. No variable in $\mathbf{V}_o \cup \mathbf{V}_m \cup \mathbf{U}$ is a child of an R unless stated. *Observed-data distribution*: $P(\mathbf{V}_o, \mathbf{V}^*, \mathbf{R})$; *underlying distribution*: $P(\mathbf{V}_o, \mathbf{V}_m, \mathbf{R})$.

8.5.4 Three categories

- **MCAR:** $\mathbf{V}_m, \mathbf{V}_o, \mathbf{U} \perp\!\!\!\perp \mathbf{R}$; no edges between $\mathbf{R}$ and $\mathbf{V}_o \cup \mathbf{V}_m$. Example: $O, A, G \perp\!\!\!\perp R_O$.
- **MAR:** $\mathbf{V}_m, \mathbf{U} \perp\!\!\!\perp \mathbf{R} \mid \mathbf{V}_o$;

  (i) no edge between any R and a partially observed variable;

  (ii) no bidirected edge between an R and a fully observed variable. Example: $A \to R_O$, $O \perp\!\!\!\perp R_O \mid A, G$.
- **MNAR:** neither. Example: $O \to R_O$, $O \not\perp\!\!\!\perp R_O \mid A, G$.

Recovering under MCAR. $P(G, O, A) = P(G, O, A \mid R_O = 0) = P(G, O^*, A \mid R_O = 0)$: list-wise deletion, which wastes data.

Recovering under MAR. $P(G, O, A) = P(G, O \mid A)P(A) = P(G, O \mid A, R_O = 0)P(A) = P(G, O^* \mid A, R_O = 0)P(A)$ (A separates (G, O) from R_O); alternatively $P(G, O, A) = P(O^* \mid A, G, R_O = 0)P(A, G)$.

MCAR implies MAR, so MAR methods apply; compare the efficiency of $P(G, O^*, A \mid R_O = 0)$ with $P(O^* \mid A, G, R_O = 0)P(A, G)$, which uses all records for $P(A, G)$.

Non-recoverability under MNAR. With $O \to R_O$, $P(G, O, A)$ is provably non-recoverable here.

A recoverable MNAR example. $Z_1 \to Z_2$, $Z_1 \to X$, $Z_1 \to Y$, $Z_2 \to X$, $X \to Y$ ($Z_2 \to Y$), Z_1 into R_{Z_2}, R_X, R_Y, $Z_2 \to R_{Z_2}$ etc. $P(Z_1) = P(Z_1 \mid R_{Z_1} = 0) = P(Z_1^* \mid R_{Z_1} = 0)$ since $R_{Z_1} \perp\!\!\!\perp Z_1$, but *not* from complete cases ($Z_1 \not\perp\!\!\!\perp R_X$, so $P(Z_1) \neq P(Z_1 \mid \mathbf{R} = 0)$). $P(Y \mid X, Z_1, Z_2) = P(Y \mid X, Z_1, Z_2, \mathbf{R} = 0) = P(Y^* \mid X^*, Z_1^*, Z_2^*, \mathbf{R} = 0)$ is recoverable from complete cases. (Proxies are often omitted.)

8.5.5 General results

Theorem 8.5 (Sequential factorization)

Given an m-graph and $P(\mathbf{V}^*, \mathbf{V}_o, \mathbf{R})$, a target Q is recoverable if it can be decomposed into an ordered (chain-rule) factorization, or a sum of such, in which every factor $Q_i = P(Y_i \mid X_i)$ satisfies $Y_i \perp\!\!\!\perp (R_{y_i}, R_{x_i}) \mid X_i$. Then each $Q_i = P(Y_i \mid X_i, R_{Y_i} = 0, R_{X_i} = 0)$.

Proof [Supplement]. Write $Q = \prod_i P(Y_i \mid X_i)$ (or a sum of such). The independence gives $P(Y_i \mid X_i) = P(Y_i \mid X_i, R_{Y_i} = 0, R_{X_i} = 0) = P(Y_i^* \mid X_i^*, R_{Y_i} = 0, R_{X_i} = 0)$ (proxies equal true values when $R = 0$), a functional of the observed distribution. ■

Theorem 8.6 (R-factorization: recoverability of $P(\mathbf{V})$)

Given an m-graph with no edges between R variables, $P(\mathbf{V})$ is recoverable iff there is no $X \in \mathbf{V}_m$ such that (1) X and R_x are neighbors, or (2) X and R_x are connected by a path whose intermediate nodes are all colliders in $\mathbf{V}_m \cup \mathbf{V}_o$. When recoverable,

$$P(\mathbf{v}) = \frac{P(\mathbf{R} = 0, \mathbf{v})}{\prod_i P(R_i = 0 \mid \mathrm{Mb}^o_{r_i}, \mathrm{Mb}^m_{r_i}, R_{\mathrm{Mb}^m_{r_i}} = 0)},$$

where $\mathrm{Mb}^o_{r_i} \subseteq \mathbf{V}_o$ and $\mathrm{Mb}^m_{r_i} \subseteq \mathbf{V}_m$ form the Markov blanket of R_i.

Proof [Supplement]. *Formula.* $P(\mathbf{v}) = P(\mathbf{v}, \mathbf{R} = 0)/P(\mathbf{R} = 0 \mid \mathbf{v})$. With no edges among the R's, $P(\mathbf{R} = 0 \mid \mathbf{v}) = \prod_i P(R_i = 0 \mid \mathbf{v}, \mathbf{R}_{-i} = 0)$, each factor equal to $P(R_i = 0 \mid \mathrm{Mb}^o_{r_i}, \mathrm{Mb}^m_{r_i}, R_{\mathrm{Mb}^m_{r_i}} = 0)$. The graphical conditions keep the variable indicated by R_i out of its Markov blanket, so each factor uses only variables observed when the relevant indicators are 0; the numerator is observed directly. *Necessity*: Mohan and Pearl (2014, 2021), not reproduced here. ■

8.5.6 Recovering a causal effect

$W \to Z \to Y$, $W \leftrightarrow Y$, $R_Y \leftrightarrow W$, Y partially observed ($Y \to Y^* \leftarrow R_Y$):

$$\begin{aligned} P(y \mid \mathrm{do}(z)) &= P(y \mid \mathrm{do}(z), r_y) = P(y^* \mid \mathrm{do}(z), r_y) \\ &= \sum_w P(y^* \mid w, \mathrm{do}(z), r_y) P(w \mid \mathrm{do}(z), r_y) = \sum_w P(y^* \mid w, z, r_y) P(w \mid r_y), \end{aligned}$$

with r_y denoting $R_Y = 0$.

Conclusion. Statisticians should model the missingness process far more often; such models have received little attention.

8.6 Measurement: general identifiability with partial observation

Motivation. Combine experiments that each measure only some variables.

Studies measure only accessible variables (oximeter for COVID vs. carbon-monoxide poisoning), so *GID-PO* uses $\mathbb{P} = \{P_{\mathbf{Z}_i}(\mathbf{V}_i)\}_i$ with $\mathbf{V}_i \subseteq \mathbf{V}$.

Example 8.4 (Exercise and stroke)

Age $A \to$ exercise X, $A \to$ BMI B, $X \to B$, $B \to$ blood pressure C, $C \to$ stroke Y, $X \leftrightarrow Y$. Given $P_X(A, C)$ (exercise experiment) and $P(B, C, Y)$ (observational):

$$\begin{aligned} P_x(y) &= \sum_{a,b,c} P_x(a, b, c, y) = \sum_{a,b,c} P(a) P_{a,x}(b) P_b(c) P_c(y) && \text{factorization} \\ &= \sum_{a,c} P(a) \Big[\sum_b P_{a,x}(b) P_b(c) \Big] P_c(y) = \sum_{a,c} P(a) P_{a,x}(c) P_c(y) && \text{proposition} \\ &= \sum_{a,c} P_x(a) P_x(c \mid a) \sum_b P(y \mid b, c) P(b) && \text{GID of each factor.} \end{aligned}$$

$P(a)P_{a,x}(c)$ comes from $P_X(A, C)$ ($P(a) = P_x(a)$, as X does not affect A); $P_c(y) = \sum_b P(y \mid b, c)P(b)$ from $P(B, C, Y)$ by adjusting for B (blocking $C \leftarrow B \leftarrow X \leftrightarrow Y$).

8.7 Conclusions

- Using abundant data requires representing the causal mechanisms, coarsely and parsimoniously.
- Necessary and sufficient conditions and algorithms exist for confounding, selection bias and generalizability (sufficient ones for missing data and partial measurement).
- Together they give a framework for pooling observational and experimental knowledge across heterogeneous domains.